

OpenAI Embeddings vs Local Model - Decision Table

Based on hands-on experiments comparing OpenAI text-embedding-3-small (1536 dims) and HuggingFace all-MiniLM-L6-v2 (384 dims) across cost, latency, privacy, and ranking quality.

Factor	OpenAI text-embedding-3-small	Local Model all-MiniLM-L6-v2	Choose	Notes
Cost	~\$0.02 per 1M tokens (pay-per-use API)	\$0.00 (runs on your hardware)	Local	For large-scale or frequent embedding, local wins clearly. OpenAI cost is negligible for small projects.
Latency	~300-800ms per batch (network round-trip)	~50-200ms per batch (CPU/GPU inference)	Local	Local is faster once the model is loaded. OpenAI latency depends on network and server load.
Data Privacy	Data sent to OpenAI servers (third-party)	Data stays on your machine (no external calls)	Local	For PII, medical, legal, or confidential data, local is mandatory. OpenAI data handling is governed by their policy.
Embedding Quality	Higher quality text-embedding-3-small: 1536 dims	Slightly lower quality all-MiniLM-L6-v2: 384 dims	OpenAI	Observed in ranking comparison: OpenAI ranked semantically closer texts higher. For precision-critical search, OpenAI is better.
Setup Complexity	API key + one SDK call (minimal setup)	Install transformers/sentence-transformers model download (~90MB)	OpenAI	OpenAI is faster to prototype. Local requires managing model versions and dependencies.
Offline / Air-gapped	Requires internet (no offline support)	Works fully offline after model download	Local	For environments without internet access (edge devices, secure networks), local is the only option.
Scalability	Scales infinitely (managed infrastructure)	Limited by your hardware (GPU helps significantly)	OpenAI	At very high volume with no GPU, OpenAI may actually be more practical despite per-token cost.
Model Updates	Automatically improved by OpenAI over time	Frozen - you control when to update	Depends	OpenAI gives you latest improvements automatically. Local gives you reproducibility - same model, same output.
Multilingual Support	Excellent (trained on diverse data)	Varies by model (MiniLM: limited)	OpenAI	For non-English text, OpenAI's models generalise much better. Multilingual local models exist but are heavier.
Production Readiness	SLA-backed, monitored rate limits apply	You manage reliability no SLA	OpenAI	For production APIs, OpenAI offloads infrastructure management. Local requires your own monitoring and failover.

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Quick-Pick Guide

Scenario	Recommendation
Regulated / sensitive data (PII, HIPAA, legal)	Local
Cost-sensitive at scale (millions of embeddings)	Local
Air-gapped or offline environment	Local
Best semantic ranking / search quality needed	OpenAI
Fast prototyping or small projects	OpenAI
High volume with no GPU available	OpenAI
Reproducibility / frozen model required	Local
Multilingual content	OpenAI